On Image Transmission Using Cooperative NOMA In VLC Network

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***Abstract*—Visible light communication uses LED lights to send information without using cables. The LED lights get brighter and darker fast, thousands of times every second. This change in brightness happens so fast that we cannot see it, but a special receiver can detect it and figure out what it means. Visible light communication works with the kind of light that we can see, which is between 380 and 780 nanometers. One good thing about light communication is that a lot of people can use the same light at the same time. There is a way to make this work called "non-orthogonal multiple access." Non-Orthogonal Multiple Access helps people who are away from the light or have a weak signal get the information they need. This means that the whole system can handle information.**

When we work together with light communication, it becomes more reliable. This helps the signal go farther and work better when things are in the way. For example, if something is blocking the light from getting to the receiver, we can still send the information. This is really helpful for people who're far away from the hospital.

In our study we sent picture information using light communication. We used a way of sending information called "on-off keying." With on-off keying, the light is either all the way on or all the way off to represent the information. Finally, we compared how well picture information is sent when we work together with light communication and when we do not, using both on-off keying and another way called maximum ratio combining. We wanted to see which way works better for sending picture information with light communication.

***Index Terms*—Index Terms—On-Off Keying (OOK), Mean Square Error (MSE), Peak Signal to Noise Ratio (PSNR), Normalized Cross Correlation (NCC), Normalized Absolute Error (NAE), Laplacian Mean Squared Error (LMSE) Demand for higher data rates has always been one of the main challenges in telecommunications.**

I. Introduction

People are using more electronics every year, and at the same time, new apps and other online services are being introduced. Due to this growth of internet traffic [1], the overall amount of wireless signals is increasing exponentially. This growth creates a problem where many wireless signals interfere with each other. Therefore, networks will reuse a common radio frequency repeatedly, and doing so only creates additional interference; consequently, there is not enough radio frequency (RF) spectrum available for all users. Also, in addition to the interference from many wireless signals, there are many limitations of the electromagnetic spectrum itself, which are limited capacity, weak security, low availability, and low efficiency.

An example of a technology that can help with these challenges is called visible light communication (VLC). VLC is a form of wireless communication that utilizes visible light from common light sources like light-emitting diodes (LEDs) or incandescent bulbs to transmit data wirelessly. The light source is modulated (flickered) at very high speeds, and the human eye cannot detect this flickering. The flickering light is received by a special photodetector that converts it back into the original data in its unmodified form. Because light is abundant in almost all indoor spaces, VLC provides a very low-cost, secure solution for transmitting data wireless without contributing to current levels of wireless radio frequency congestion.

A very useful idea for sending pictures using light. Pictures have been one of the most popular methods that people use today to share information. For example, at business meetings and in teaching classrooms using photographs or during storytelling and creating digital art, photographs have become an important way for us all to communicate. An On-Off Keying (OOK) modulation method is used in VLC systems. OOK MODULATION uses a flash of light to represent a binary "1" and no light to represent a binary "0." This is a straightforward technique for converting digital data into light signals and from light signals back to digital data.

Enhancing cooperative communication will improve wireless network communications' capacity, speed, and overall performance. Cooperative communication will also extend the communication distance of both mobile networks and ad-hoc networks. Decode-and-forward relay (DFR) communication will be used as the principal technique in this article. In DFR communication, a middle user assists in relaying the signal, or message, to the final destination. The middle user has three functions in this process: receiving the received signal, decoding the received signal, and re-sending the re-encoded signal to the final destination. Thus, with DFR communication, the reliability of communication in VLC systems will be enhanced, particularly when the end user is located a great distance from the original transmitter and/or when the light has been obstructed by walls, furniture, and other objects.

Research published until now on cooperative VLC systems has focused on cooperative VLC systems in general. Nevertheless, only two of the wide range of papers published to date have investigated how cooperative VLC systems work for sending images. One paper discusses a hybrid approach where

RF is used with VLC to provide a mechanism for cooperative communications between vehicles. The other discusses a cooperative VLC system that uses On-Off Keying (OOK) modulation in both ideal and non-ideal channel conditions. The other two papers are as follows: One proposes a relaying strategy whereby the relay receives the image data via VLC from the sender and transmits the image data to the destination via RF; and the other presents a NOMA (Non-Orthogonal Multiple Access)-based cooperative VLC system that combines OOK and L-Pulse Position Modulation to relay images to the destination. Researchers have also examined MIMO-VLC systems with receivers oriented at various angles, as well as the impact of energy consumption and alternative modulations on the potential for signal outage in systems employing selective/incremental Decode and Forward (DF) relaying systems.

The authors of this paper contend that there has not been any previous research that compares cooperative (i.e., cooperatively relaying the image data) and non-cooperative (i.e., sending the image data directly) VLC systems using OOK modulation and Maximum Rank Communication (MRC), and thus, this paper fills that research gap.

The layout of the final portion of this document is kept simple. Section 2 goes into detail about the OOK based VLC system as well as system and channel models, and the packet structure for signals. Section 3 discusses performance evaluations used in comparison of non-cooperative versus cooperative communication, including metrics such as MSE and PSNR, etc. Section 4 presents experimental results with discussion. Section 5 concludes with a recap of the findings and provides suggestions for future research.

II. COLLABORATIVE DATA TRANSMISSION IN VLC NETWORKS

A. System Configuration and Geometric Arrangement

In an indoor visible light communication system, we have one node that sends information and two nodes that receive it. One of these receiving nodes helps the other one get the information and the other one is the destination. The node that sends information has one emitting diode or LED and it is placed in the middle of the ceiling of a room that is 8 meters long 8 meters wide and 4 meters high. The light from this LED goes out in a pattern and this pattern is important for the nodes that receive the information.

The node that helps the one, called U1 has two main parts. One part receives the information. It has a special device called a photodetector or PD. This device looks at the light. Figures out what the message says. Then it sends the message to the part of U1, which sends it out again as light. This light has a pattern and it is strong enough to reach the final destination.

When we use Light Communication in a big room it is helpful to have nodes that can help each other. This is because the light can only go far and it can be blocked by things in the room. So we use a system where one node helps another node get the information. We can see a picture of this system in Figure 1 where two nodes, U1 and U2 work together to get the information.

When we send information there are two steps. The first step is called the broadcast step. In this step the node that sends information called Tx sends the message to all the nodes. The second step is called the cooperation step. In this step U1 helps U2 get the message. U1 sends the message again. U2 gets two messages, one from Tx and one, from U1. Then U2 uses a technique to combine these two messages and get the information it needs.

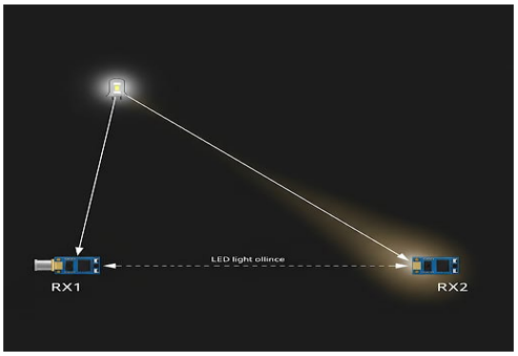


Fig. 1. System Model for 2 User Cooperative VLC System

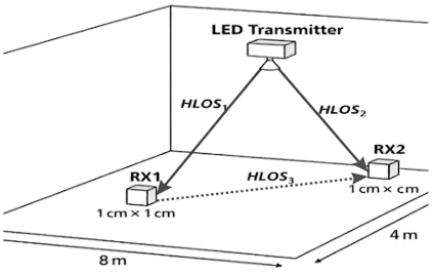


Fig. 2. Room Setup for VLC System

The line-of-sight (LOS) channel impulse response (CIR) is given by[14]

$$H_{LOS} = \begin{cases} \frac{A}{D_{i,j}^2} R_\ell(\psi_i)\, G_c(\phi_{i,j})\, G_s(\phi_{i,j}) \cos(\phi_{i,j}), & 0 \le \phi \le \Phi_c \\ 0, & \phi > \Phi_c \end{cases} \tag{1}$$

where A represents the physical area of the photodetector, $D_{i,j}$ represents the Euclidean distance between the ith transmitter and j^{th} receiver, ϕ represents the angle of incidence, Φ_c represents the field of view (FoV) of Pphto-Detector, and ψ represents the irradiance angle. $G_c(\phi_{i,j})$ represents the gain of optical filter,

The gain of an optical concentrator is given by:

$$G_s(\phi_{i,j}) = \frac{n^2}{\sin^2(\Phi_c)}, \quad 0 \le \phi \le \Phi_c \tag{2}$$

Where n is the refractive index of the optical concentrator. The Lambertian radiant intensity is given by:

$$R_\ell(\psi_i) = \frac{m+1}{2\pi}\cos^n(\psi_i) \tag{3}$$

where n represents the order of Lambertian emission defined as:

$$m = \frac{-\ln(2)}{\ln(\cos(\psi_{1/2}))} \tag{4}$$

represents the order of Lambertian emission represented by semi-angle at half power of an LED [15].

B. Signal Structure

The transmitted signal can be expressed as $s = 2P_t b$, where b represents the transmitted symbol and P_t denotes the average transmitted power. The signal received at the kth receiver from the lth transmitter is given by

$$r_k = 2\rho H_{LOS} P_t b + w \tag{5}$$

where H_{LOS} represents the LOS channel gain between the lth transmitter and k^{th} receiver, ρ is the responsivity of the photodetector, and w denotes additive white Gaussian noise (AWGN).

The noise variance is given by

$$\sigma_w^2 = \sigma_{th}^2 + \sigma_{sh}^2 \tag{6}$$

where σ_{th}^2 and σ_{sh}^2 compares the received signal against a threshold, and then uses the decode-and-forward (DF) method to relay the detected signal to the destination, D_1.

The destination will receive both the signal sent directly by the source and the signal sent by the relay. The two signals will be combined using maximum ratio combining (MRC) prior to being combined.

Prior to combining the two signals, each signal will be assigned a weighting factor based on the quality of the signal. This will result in an improved overall signal-to-noise ratio (SNR) and an improved quality of reception [8].

TABLE I
SIMULATION PARAMETERS

Description	Value
Size of Room	$8\,m \times 8\,m \times 4\,m$
Source Coordinates	$4\,m \times 4\,m \times 4\,m$
U_1 Coordinates	$7\,m \times 6\,m \times 1.5\,m$
U_2 Coordinates	$4\,m \times 4\,m \times 1.5\,m$
Semi-half angle of LED ($\psi_{1/2}$)	60°
FoV (Φ_c)	60°
Physical Area of PD (A)	$1\,\text{cm}^2$
Refractive index (n)	1.5
Gain of optical filter $G_c(\phi_{i,j})$	1
Transmission rate (R_b)	50 Mbps
Noise bandwidth (B)	50 MHz
Background Current (I_B)	$202\,\mu A$

III. PERFORMANCE PARAMETERS

The performance of the proposed cooperative VLC-NOMA system is evaluated using multiple image quality and error metrics, including MSE, PSNR, NCC, NAE, LMSE, and BER. These parameters are analyzed with respect to varying SNR values to understand the effectiveness of cooperative transmission.

A. Mean Square Error (MSE)

MSE allows you to determine how closely an original image matches its reconstructed form based on the average squared distance between the differing pixels of the images. The lower the MSE value is, the more accurate the reconstructed image will be to the original image with minimal visual errors/distortions.

$$MSE = \frac{1}{P \times Q}\sum_{p=1}^{P}\sum_{q=1}^{Q}\left[I(p,q) - \hat{I}(p,q)\right]^2 \tag{7}$$

where, P and Q are the dimensions (width and height) of the images. $I(p,q)$ and $\hat{I}(p,q)$ represent the pixel intensity values of the original and reconstructed images, respectively, at position (p,q).

B. Peak Signal to Noise Ratio (PSNR))

PSNR is a metric used to compare a reconstructed image against an original image to determine quality. Like MSE, PSNR is expressed in decibels (dB) and functions oppositely of MSE; when MSE values decrease (indicating less distortion), corresponding PSNR values will increase (indicating an improved match). Thus, a higher PSNR indicates fewer defects in the reconstructed image.

$$PSNR = 10\log_{10}\left(\frac{I_{max}^2}{MSE}\right) \tag{8}$$

where, I_{max} represents the maximum possible pixel value (e.g., 255 for 8-bit images).

C. Normalized Cross Correlation (NCC)

NCC is a method used to compare two images by comparing the relative alignment of their corresponding pixel patterns. The NCC score ranges between -1 and 1; thus, the higher the NCC score is (i.e., the closer the NCC score is to 1) indicates the reconstruction's legs are more similar to those of the original image.

$$NCC = \frac{\sum_{p=1}^{P}\sum_{q=1}^{Q}(I(p,q) - \bar{I})(\hat{I}(p,q) - \bar{\hat{I}})}{\sigma_I \cdot \sigma_{\hat{I}}} \tag{9}$$

where, $\bar{I}$ and $\bar{\hat{I}}$ are the mean pixel values, while σ_I and $\sigma_{\hat{I}}$ are the standard deviations of the original and reconstructed images, respectively.

D. Normalized Absolute Error (NAE)

Normalized Absolute Error (NAE) examines the reconstruction accuracy of an image by dividing the total absolute differences (pixel by pixel) between the identical pixel values and the lowest possible pixel value by the percentage difference, using the entire possible range of pixel values as the denominator. This ensures that the amount of overall brightness/darkness does not affect the NAE calculation, resulting in a smaller NAE for improved reconstruction accuracy and decreased distortion.

$$NAE = \frac{1}{P \times Q \times L} \sum_{p=1}^{P} \sum_{q=1}^{Q} \left| I(p,q) - \hat{I}(p,q) \right|^2 \quad (10)$$

where, P and Q represent the dimensions of the image, L denotes the maximum possible pixel value (e.g., 255 for 8-bit images), and $I(p,q)$ and $\hat{I}(p,q)$ represent the original and reconstructed pixel values at position (p,q).

E. Laplacian Mean Squared Error (LMSE)

The LMSE metric compares the quality of an image by first applying a Laplacian filter to both original and reconstructed images to detect edge and fine-detail increases. The LMSE metric then compares the two Laplacian filtered images and reports an LMSE value. The lower the LMSE value is, the more successful the reconstructed image will be in preserving edge and small-detail information.

$$ALMSE = \frac{1}{P \times Q} \sum_{p=1}^{P} \sum_{q=1}^{Q} \left[L(I(p,q)) - L(\hat{I}(p,q)) \right]^2 \quad (11)$$

where, $L(I(p,q))$ and $L(\hat{I}(p,q))$ represent the Laplacian-transformed intensities of the original and reconstructed images, respectively.

IV. RESULT AND ANALYSIS

The performance of the proposed VLC system is evaluated under both cooperative and non-cooperative transmission schemes. The analysis is carried out over a range of Signal-to-Noise Ratio (SNR) values to examine the behavior of different performance metrics such as MSE, PSNR, NCC, NAE, and LMSE.

Fig. 3 illustrates the variation of Mean Square Error (MSE) with respect to SNR. It is observed that the MSE decreases as the SNR increases, indicating improved reconstruction of the transmitted image. Moreover, the cooperative system consistently exhibits lower MSE compared to the non-cooperative system, which confirms reduced distortion and better accuracy.

Fig. 4 shows the PSNR performance. It can be seen that PSNR increases with increasing SNR, implying better image quality at higher signal strengths. The cooperative approach achieves higher PSNR values than the non-cooperative case, demonstrating enhanced visual quality of the received image.

The variation of Normalized Cross-Correlation (NCC) is depicted in Fig. 5 The NCC values move closer to unity as SNR increases, indicating higher similarity between the original and reconstructed images. The cooperative transmission yields higher NCC values at all SNR levels, confirming better structural similarity.

Fig. 6 presents the Normalized Absolute Error (NAE) analysis. The NAE decreases with increasing SNR, showing a reduction in pixel-wise error. It is evident that the cooperative system produces lower NAE values compared to the non-cooperative system, indicating improved accuracy in image reconstruction.

Fig. 7 shows the LMSE performance, which focuses on edge preservation and detail accuracy. The LMSE decreases as SNR increases, and the cooperative system consistently outperforms the non-cooperative system by producing lower LMSE values.

Fig. 8 shows that the Bit Error Rate (BER) decreases as the SNR increases, indicating improved communication reliability. The cooperative system achieves lower BER than the non-cooperative system due to relay-assisted transmission and MRC combining, resulting in better error performance.

V. IMAGE ANALYSIS

To visually demonstrate the improvement achieved by incor-porating the cooperative NOMA, this section presents the reconstructed images at the receiver for four cases: the first original image used for transmission,the Second original image used for transmission, the image received at user Rx1 , the image received at user Rx2. The comparison clearly highlights the benefits of indicating better structural similarity , improved reconstruction of the transmitted image.

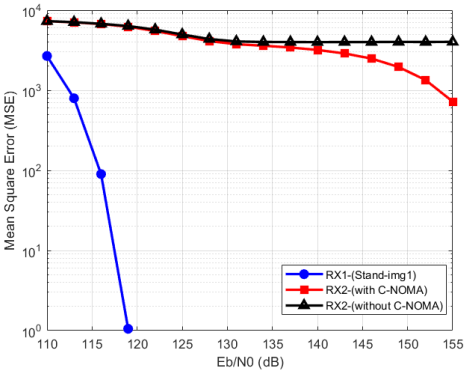


Fig. 3. MSE VS SNR

TABLE II
PERFORMANCE ANALYSIS OF VLC SYSTEM

P1	MSE		PSNR (dB)		NCC		NAE		LMSE	
	Rx1	Rx2	Rx1	Rx2	Rx1	Rx2	Rx1	Rx2	Rx1	Rx2
0.5	9328.74	9328.74	8.43	8.43	0.8133	0.9204	0.5891	0.6871	2.0290	72.0951
0.6	9328.74	9328.74	8.43	8.43	0.8133	0.9204	0.5891	0.6871	2.0290	72.0951
0.7	0.0000	9328.74	204.67	8.43	1.0000	0.9204	0.0000	0.6871	0.0000	72.0951
0.8	0.0000	83.2715	204.67	28.93	1.0000	0.9970	0.0000	0.0085	0.0000	10.8892
0.9	0.0000	0.0000	204.67	204.67	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000

VI. CONCLUSION

In this paper, the performance of image transmission over a cooperative VLC-NOMA system has been analyzed. The

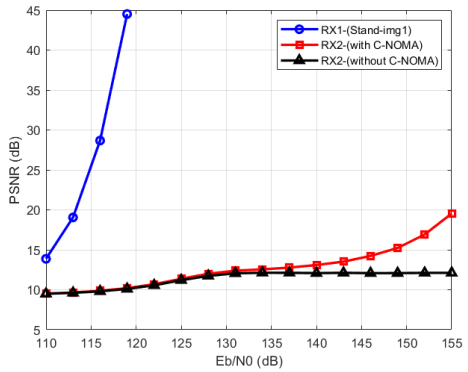


Fig. 4. PSNR VS SNR

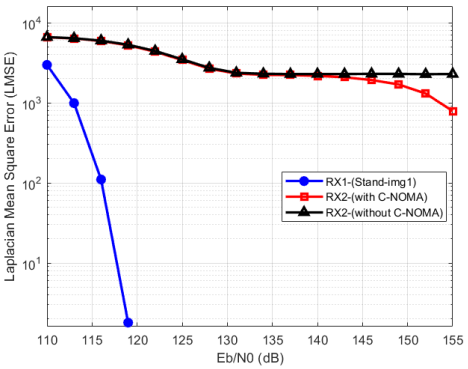


Fig. 7. LMSE VS SNR

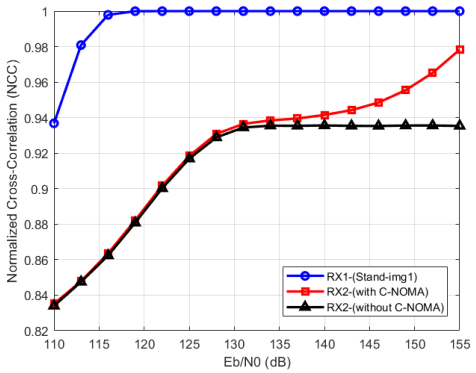


Fig. 5. NCC VS SNR

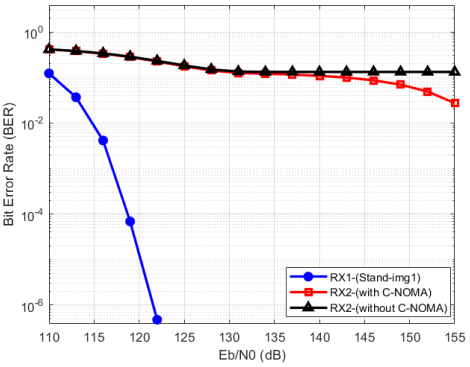


Fig. 8. BER VS SNR

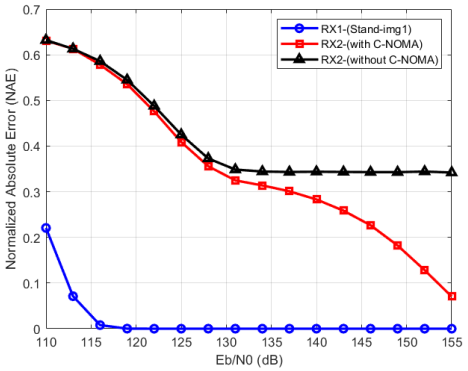


Fig. 6. NAE VS SNR



Fig. 9. Input image1 used for VLC transmission



Fig. 10. Input image2 used for VLC transmission



Fig. 11. Output Image at RX1



Fig. 12. Output Image at RX2

system utilizes OOK modulation along with decode-and-forward (DF) relaying and maximum ratio combining (MRC) to improve communication reliability.

The performance evaluation is carried out using different image quality metrics such as MSE, PSNR, NCC, NAE, LMSE, and BER. The simulation results show that the co-operative transmission scheme significantly outperforms the non-cooperative approach. The MSE is reduced, PSNR is increased, and NCC values are closer to unity, indicating better image reconstruction quality.

Furthermore, the BER performance also improves with cooperative transmission due to the presence of relay-assisted communication, which enhances the effective SNR and reduces transmission errors.

Overall, the results demonstrate that cooperative communication plays a crucial role in improving the reliability, efficiency, and quality of image transmission in VLC systems, especially for users located far from the transmitter.

Future work may include the implementation of advanced modulation techniques, adaptive power allocation, and deep learning-based decoding methods to further enhance system performance.

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